Software Requirement Specifications

AI Career & Skill Guidance System

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Summary

The **AI Career & Skill Guidance System** is an intelligent advisory platform designed to help students and job seekers choose the most suitable career path based on their interests, skills, academic background, and long-term goals. Many students face difficulty understanding which field aligns with their strengths or what skills are required to succeed in their desired career. This system solves that problem using AI-powered recommendations, skill-gap analysis, curated learning resources, and career comparison features.

The system integrates modern AI tools, automation workflows, and light coding components to create a robust yet easy-to-maintain solution. It provides high-quality learning links, visual progress dashboards, voice-based interaction, and automated data processing. This SRS document outlines the complete structure, requirements, constraints, and system behavior following an IEEE-based format to guide development effectively.

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1. Introduction

The **AI Career & Skill Guidance System** is designed to address one of the most common challenges faced by students and job seekers today—selecting the right career path and understanding which skills are required to succeed in that field. With thousands of career options and rapidly evolving skill requirements, many learners struggle to make informed decisions. This system uses Artificial Intelligence, automation tools, and structured data analysis to provide personalized recommendations based on user interests, skills, academic background, and future goals. The system aims to simplify career planning by delivering AI-powered guidance through an interactive chatbot interface, resource suggestions, career comparisons, and progress tracking dashboards. This introduction provides an overview of the purpose, scope, users, literature/background, and technologies proposed for the development of the system.

1.1. Purpose

The purpose of the AI Career & Skill Guidance System is to provide an intelligent advisory platform that assists users in identifying suitable career options and the skills required to excel in those careers. The system aims to solve the confusion faced by students regarding future career choices by delivering AI-generated recommendations based on their personal profile. It also offers a structured learning roadmap, skill-gap analysis, voice-based interaction, and automated progress tracking. This SRS document defines all technical and non-technical requirements, ensuring that developers and evaluators clearly understand the system’s goals and functionality.

1.2. Scope

The scope of this project includes designing and developing a web-based AI-powered recommendation system that performs the following functions:

- Collects user inputs such as interests, skills, academic field, and goals
- Generates personalized career recommendations
- Provides skill-gap analysis for chosen career paths
- Suggests relevant courses and learning resources
- Allows comparison between multiple career options
- Tracks user learning progress through dashboards
- Offers chatbot and voice interaction features
- Stores and updates data using cloud-based tools

This project **will not** include job application automation, real-time job scraping, or integration with paid APIs beyond prototype limits. The system focuses on career guidance, not job placement.

1.3. Product Perspective

This system is a standalone advisory module that does not rely on any existing institutional system. It integrates various external tools and services including:

- AI Models (OpenAI GPT)
- Voiceflow for chatbot flow
- Make.com for workflow automation
- Google Sheets for data storage
- Looker Studio for dashboards
- APIs such as YouTube Data API and Coursera links

Light coding components (HTML, CSS, JSON-based logic, and API integration) are used to enhance customization and improve system flexibility. The system can be scaled in the future by adding more careers, APIs, and learning platforms.

1.4. User Characteristics

The system is designed for a wide range of users with basic computer or smartphone usage skills. The primary user categories include:

1.4.1 Students:

Students at school, college, or university level who need help identifying the right career path.

1.4.2 Job Seekers:

Individuals looking to upskill or transition into better job roles.

1.4.3 Academic Advisors:

Teachers or counselors who may use the system to guide students.

1.4.4 Beginners:

Users with limited technical knowledge who need structured guidance and easy interaction tools.

No advanced technical knowledge is required to operate the system.

1.5. Similar apps and systems/Literature Review

In Several career guidance platforms exist today, such as:

a) Google Career Certificates

Offers skill-based courses but lacks personalized AI recommendations.

b) Coursera Career Academy

Provides structured pathways but requires paid enrollment and does not perform personalized skill-gap analysis.

c) LinkedIn Learning Career Explorer

Shows job trends and skills but does not offer customized for-student recommendations or voice interaction.

d) ChatGPT-based unofficial career advisors

Provide recommendations but lack dashboards, tracking, workflows, or career comparison.

Gaps Identified in existing systems:

- Limited personalization
- No skill progress tracking
- No voice interaction
- No automated workflows
- No consolidated comparison between careers

Our system fills these gaps by providing AI-powered personalized guidance, dashboards, comparisons, voice mode, and automation.

1.6. Proposed Technologies

AThe following technologies and platforms are proposed for the development of the system:

- **Voiceflow** – Chatbot and conversation design
- **OpenAI GPT API** – AI reasoning and recommendations
- **Make.com / Zapier** – Automation workflows
- **Google Sheets** – Data storage
- **Google Looker Studio** – Dashboard and progress tracking
- **HTML, CSS** – Basic front-end customization
- **JSON Logic** – Custom data handling and AI structure
- **YouTube Data API** – Fetch curated learning resources
- **Coursera/Udemy/FreeCodeCamp Links** – Course recommendations
- **Netlify** – Deployment of UI components

These technologies ensure system flexibility, scalability, easy maintenance, and professional execution.

2. Requirements

The AI Career & Skill Guidance System consists of several major functions that help users receive personalized career recommendations. The system collects user information such as interests, educational background, and skills, and analyzes it using AI. It identifies skill gaps, suggests learning resources, provides career comparisons, and tracks user progress. These functions work together to guide students and job seekers toward suitable career paths with accurate and structured recommendations.

2.1. Function Requirements

Functional requirements define the specific actions and behaviors that the system must perform. These requirements describe how the user interacts with the system and how the system processes, stores, and responds to the information. The following requirements describe the detailed functionalities of the AI Career & Skill Guidance System, including user AI-based career recommendations, skill gap analysis, course suggestions, comparison features, and progress tracking.

2.1.1 User Registration & Input Collection

FR-001: User Profile Input

The system shall allow the user to provide personal details including name, field of study, interests, strengths, and learning goals.

FR-002: Skill Input

The system shall allow users to enter their existing skills for better matching.

FR-003: Data Validation

The system shall validate user inputs to ensure data accuracy (e.g., empty fields, invalid formats).

2.1.2 Career Recommendation Module

FR-004: AI Recommendation Generation

The system shall generate personalized career recommendations using AI analysis based on user inputs.

FR-005: Priority Ranking

The system shall rank career recommendations based on suitability, interest alignment, and skill matching scores.

FR-006: Detailed Career Insights

The system shall provide insights for each recommended career including required skills, job scope, and learning difficulty.

2.1.3 Skill Gap Analysis

FR-007: Missing Skill Identification

The system shall identify missing or weak skills based on the user's chosen career.

FR-008: Skill Level Mapping

The system shall map user skills to the expected industry-level skills.

2.1.4 Learning Resource Recommendation

FR-009: Course Link Suggestions

The system shall provide learning links from YouTube, Coursera, Udemy, Google Digital Garage, and FreeCodeCamp.

FR-010: Resource Categorization

The system shall categorize resources such as beginner, intermediate, and advanced.

FR-011: AI-Based Course Filtering

The system shall filter out irrelevant resources and show only the most suitable learning materials.

2.1.5 Career Comparison Tool

FR-012: Comparison Report Generation

The system shall allow users to compare two or more careers based on

- required skills
- salary expectations
- future demand
- learning time
- difficulty level

FR-013: Visual Comparison Output

The system shall generate a visually structured comparison chart using Google Sheets dashboard.

2.1.6 Progress Tracking Module

FR-014: Progress Data Storage

The system shall store user learning progress in Google Sheets.

FR-015: Progress Visualization

The system shall generate charts/graphs for user progress through Looker Studio dashboards.

FR-016: Module Completion Updates

The system shall allow users to update completed learning modules.

2.1.7 Voice & Chat Interaction

FR-017: Voice Interaction Mode

The system shall allow the user to interact with the system using voice commands through Voiceflow.

FR-018: Chat-based Guidance System

The system shall provide interactive chat conversation powered by AI assistance to guide users step-by-step.

2.1.8 Automated Workflow Functionality

FR-019: Automated Data Processing

The system shall automatically transfer user input and results to Google Sheets via Make.com workflows.

FR-020: Report Automation

The system shall automatically generate personalized recommendation reports using automation tools.

2.1.9 System Output Generation

FR-021: Final Recommendation Report

The system shall generate a structured summary for the user including:

- top careers
- learning roadmap
- suggested courses
- comparison insights
- progress summary

2.2. Non-Functional Requirements

Non-functional requirements define the quality attributes, performance, and standards that the AI Career & Skill Guidance System must maintain to operate efficiently and provide a smooth user experience.

🔹 1. Performance Requirements

- The system should respond to user queries within **2–4 seconds**.
- AI recommendation generation should not exceed **7 seconds**, even during high traffic.
- The system must support at least **1000 concurrent users** without performance degradation.
- Data retrieval from Google Sheets/Database should be processed in **real-time**.

🔹 2. Security Requirements

- User passwords must be stored using **encrypted hashing mechanisms**.
- Only authenticated users should access recommendation results and personal profiles.
- API keys (OpenAI, Zapier, Make.com, etc.) must be securely stored and not visible in frontend.
- System should verify user identity during password reset via **email verification**.
- Role-based access should be implemented if an Admin panel is added later.

🔹 3. Usability Requirements

- The user interface must be simple, clear and easily understandable for students.

- All features (sign up, profile update, career recommendation) should be accessible within **3 clicks**.
- System must support **mobile-friendly** view for Android/iOS users.
- Buttons, labels, and instructions must be written in **simple English** for easy navigation.

🔹 4. Reliability Requirements

- The system should remain available **99% of the time**.
- User data (skills, progress, interests) must be backed up automatically.
- If an API fails (OpenAI, YouTube, Coursera API), the system must display a friendly fallback message.
- System must run consistently without unexpected errors or crashes.

🔹 5. Maintainability Requirements

- System should be easy to update using low-code/no-code tools (Zapier, Make, Voiceflow).
- Admin should be able to update career data, skills, and course lists without coding.
- Documentation (README + workflow notes) must be maintained for future developers/students.
- Code/workflows should be structured so changes can be made within **15–20 minutes**.

🔹 6. Scalability Requirements

- System must support increasing users without redesigning the entire system.
- New careers, skills, and APIs (Udemy, edX, Google Garage) should be easily addable.
- The architecture should allow integration of new modules like job alerts or internship finder in future.

🔹 7. Compatibility Requirements

- System must run smoothly on: Chrome, Firefox, Edge.
- Should support devices: desktop, laptop, tablet, mobile.
- Workflows must integrate cleanly with OpenAI, Google Sheets, Zapier, Make.com, and Voiceflow.

🔹 8. Data Integrity Requirements

- User data must remain accurate during storage, processing, and updating.
- Duplicate user accounts should be prevented using email validation.
- Progress tracking data must remain correct even after multiple updates.

🔹 9. Accessibility Requirements

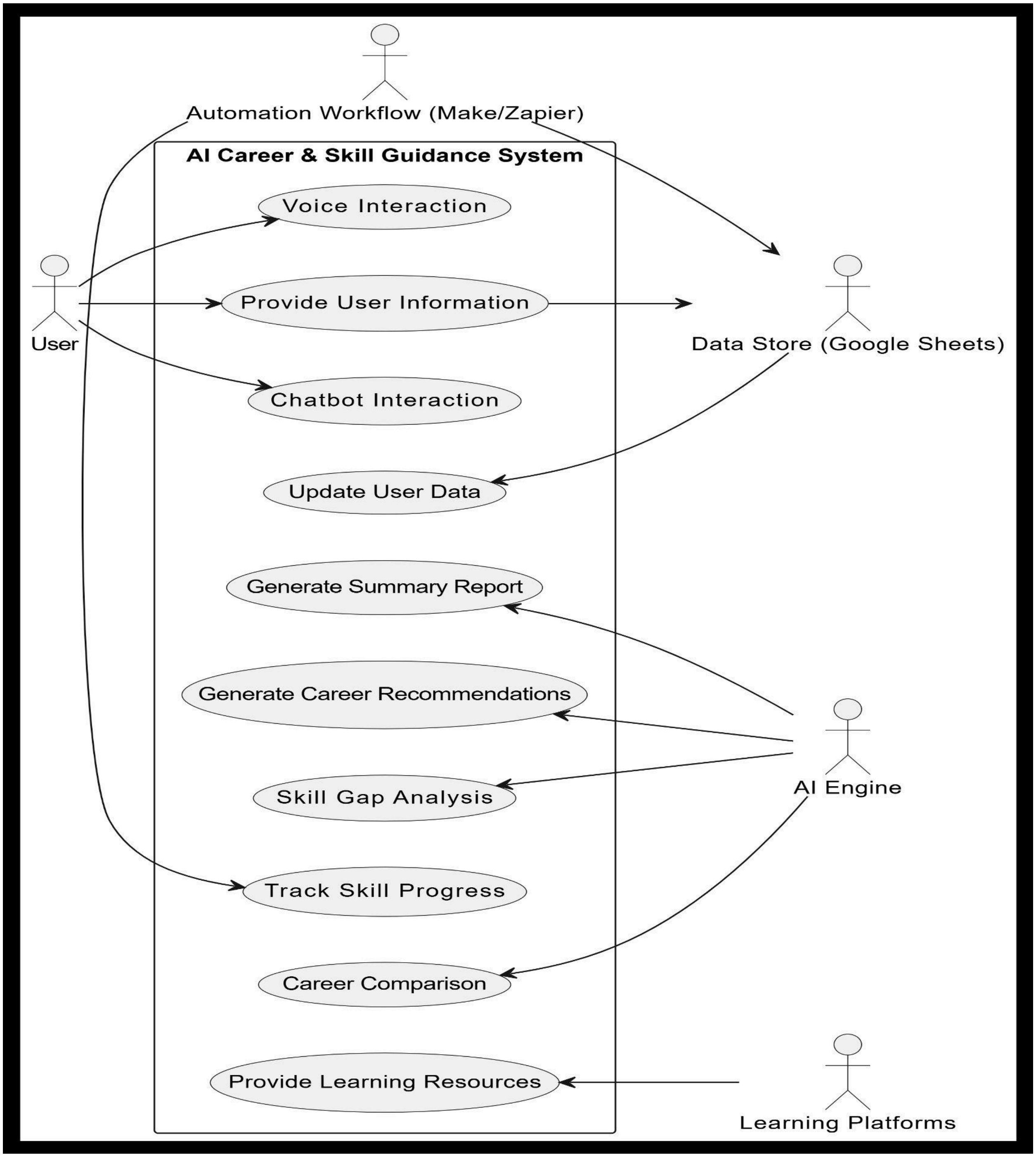
- System should support screen-friendly colors for readability.

- Voice interaction feature should assist users who prefer speaking instead of typing.
- Text content should follow minimum **14px readable font standards**.

3. Use Cases and Flow of Processes

The *Use Case and Flow of Processes* section provides a detailed explanation of how users and system components interact with the **AI Career & Skill Guidance System**. It identifies all major actors, their goals, and the functional steps involved in achieving those goals. This section outlines each use case with structured descriptions, helping to clarify the operational behavior of the system. Additionally, the process flow illustrates how user input moves through different modules—such as AI recommendation, skill-gap analysis, course suggestion engine, comparison module, automation workflows, and progress tracking dashboards—to produce meaningful outputs. Together, these use cases and process flows present a complete picture of the system's functionality and ensure that all requirements are covered through clear, logical, and traceable interactions.

USE CASE DIAGRAM:



System Level Use Case Diagram

Use Case 1 : Provide User Information

Attribute	Description
Use Case ID	UC1
Use Case Name	Provide User Information
Actor	User
Description	User enters personal details, interests, skills, and academic background required for personalized analysis.
Preconditions	User must be connected to the system/chatbot.
Postconditions	User data is stored in Google Sheets for further processing.
Main Flow	1. System asks user questions → 2. User provides answers → 3. Data sent to backend workflow → 4. Stored in Google Sheets
Alternate Flow	User skips a question → System asks for clarification or continues.

Use Case 2 : Generate Career Recommendations

Attribute	Description
Use Case ID	UC2
Actor	AI Recommendation Engine
Description	System analyzes user input using AI (OpenAI GPT) and generates personalized career suggestions.

Preconditions	Required user data must be available.
Postconditions	Suitable careers are displayed to the user.
Main Flow	Data ⟶ AI engine ⟶ Reasoning prompt ⟶ Final career list ⟶ Output to chatbot
Alternate Flow	Insufficient data ⟶ System requests missing info.

Use Case 3: Skill Gap Analysis

Attribute	Description
Use Case ID	UC3
Actor	AI Engine
Description	System compares user skills with required skills for selected career and identifies gaps.
Preconditions	Career must be selected.
Postconditions	Missing skills are displayed.
Main Flow	User selects career ⟶ AI analyzes skill list ⟶ Output missing skills

Use Case 4: Provide Learning Resources

Attribute	Description
Use Case ID	UC4

Actor	AI System + Learning Platforms
Description	System recommends learning materials from YouTube, Coursera, FreeCodeCamp, and Udemy.
Preconditions	Skill gap analysis completed.
Postconditions	User receives course links.
Main Flow	AI selects platform → Generates list → Sends through chatbot

Use Case 5: Career Comparison

Attribute	Description
Use Case ID	UC5
Actor	User, AI Engine
Description	User compares two or more careers on the basis of difficulty, skills, salary, growth, learning duration, and demand.
Preconditions	User selects careers for comparison.
Postconditions	Comparison chart displayed.

Use Case 6: Track Skill Progress

Attribute	Description
Use Case ID	UC6
Actor	User + Automation System

Description	System tracks completed skills and displays progress on Google Looker Studio dashboard.
Preconditions	User's skill updates exist.
Postconditions	Updated progress visualization.

Use Case 7: Update User Data Automatically

Attribute	Description
Use Case ID	UC7
Actor	Make.com / Zapier
Description	System automatically stores and updates user data without manual entry.
Preconditions	Workflow connected.
Postconditions	Data synced with sheets & dashboard.

Use Case 8: Generate Report

Attribute	Description
Use Case ID	UC8
Actor	System
Description	System creates summary including AI recommendations, skill gaps, resources, and comparison.

Use Case 9: Voice Interaction

Attribute	Description
Use Case ID	UC9
Actor	User + Voiceflow
Description	User interacts with system using voice mode.

Use Case 10: Chatbot Interaction

Attribute	Description
Use Case ID	UC10
Actor	User
Description	Chatbot guides user through step-by-step conversation.

FLOW OF PROCESSES:

🔹 **Step 1 – User Starts Interaction**

User opens the chatbot interface. System welcomes user and starts asking questions.

🔹 **Step 2 – User Profile Collection**

User provides required information:

- Name
- Academic field
- Skills
- Interests
- Future goals

  Data automatically goes to Google Sheets.

🔹 Step 3 – AI Processing

OpenAI GPT recommendation engine analyzes the user profile and generates:

- Suitable careers
- Reasoning behind selections

Automation workflow sends data to the AI engine.

🔹 Step 4 – Skill Gap Analysis

System compares user's current skills with required skills of career options.

🔹 Step 5 – Course Recommendation Process

Using AI + API logic, system selects high-quality learning materials.

🔹 Step 6 – Career Comparison Module

System creates comparison table including:

- Difficulty
- Salary

- Growth rate
- Skill requirements
- Learning duration

🔹 Step 7 – Progress Tracking Workflow

User skill updates are stored → Dashboard auto-updates.

🔹 Step 8 – Report Generation

System generates:

- Career list
- Skill roadmap
- Course links
- Comparison chart
- Progress report

🔹 Step 9 – Voice Interaction (Optional)

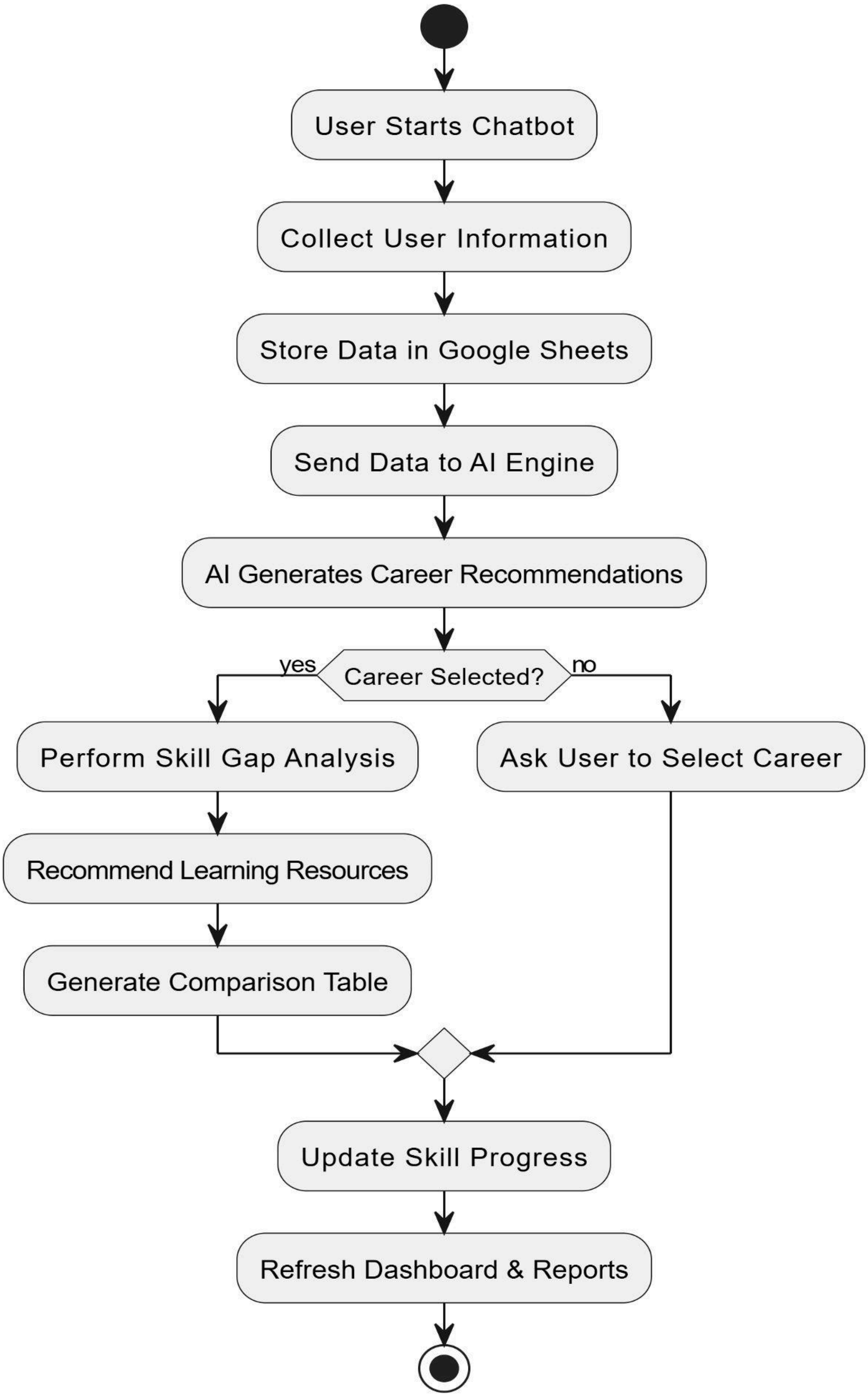
User can enable voice mode for communication.

🔹 Step 10 – System Ends Session

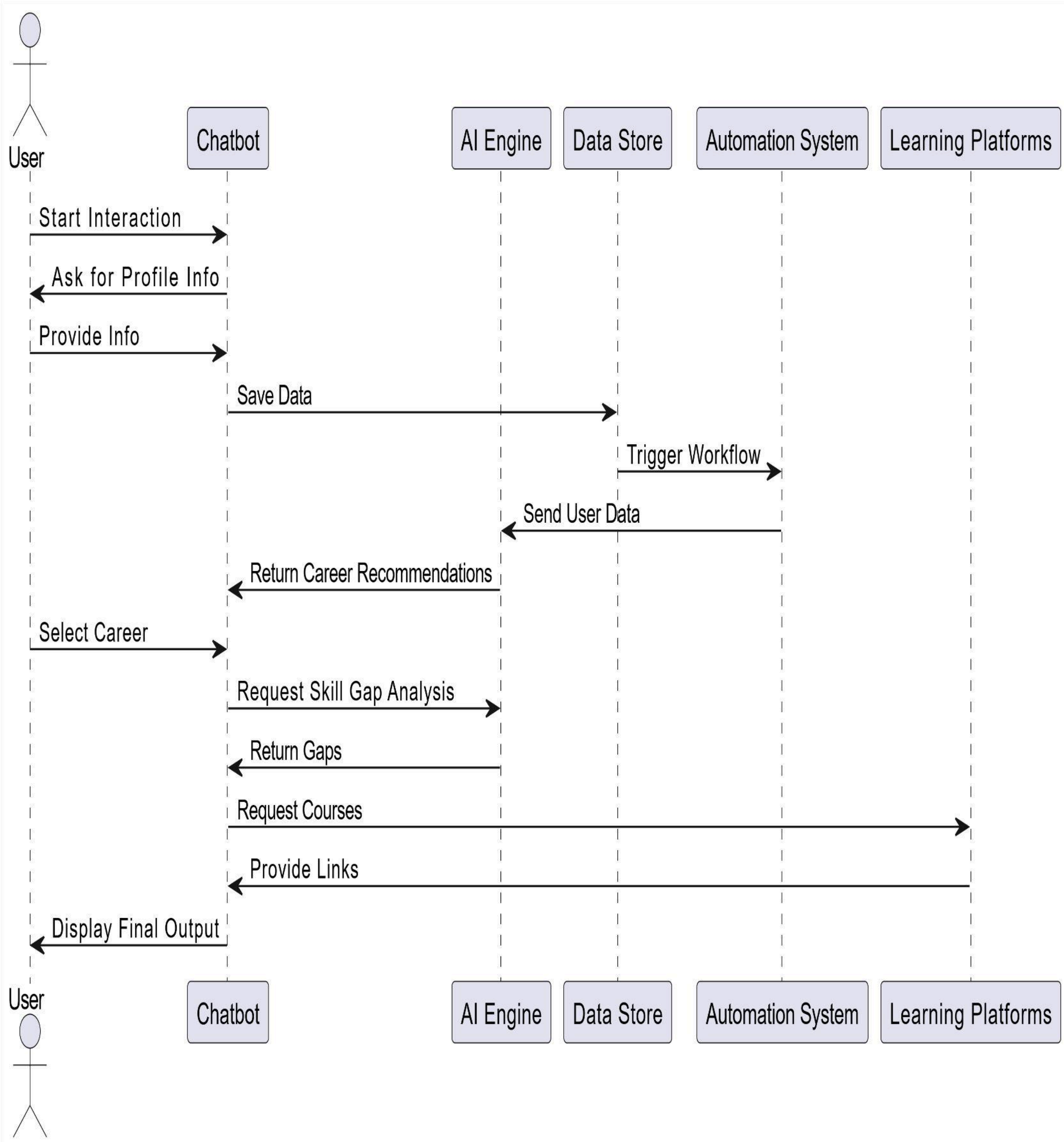
Outputs summary and ends flow.

Diagrams:

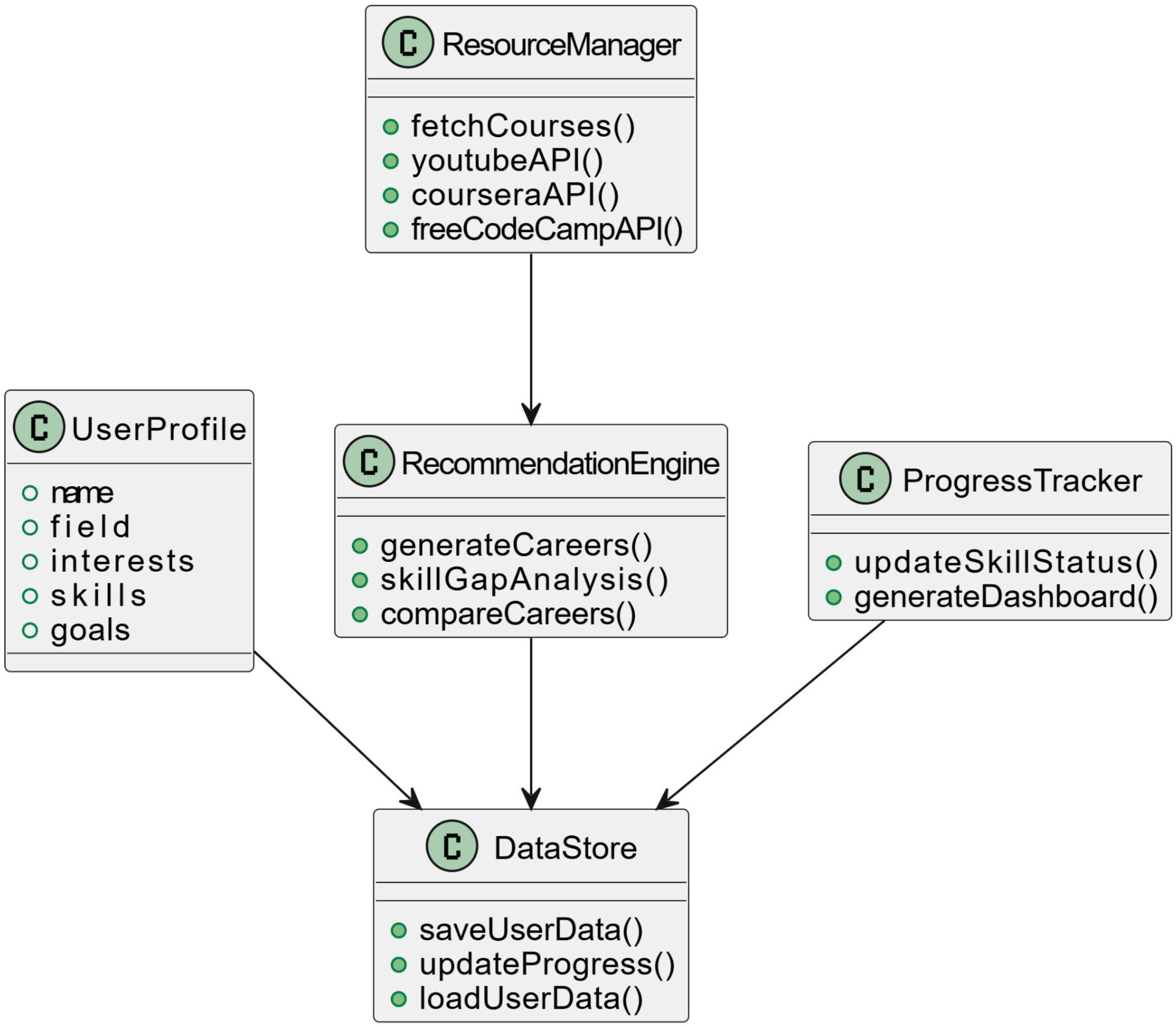
✅ ACTIVITY DIAGRAM – SYSTEM WORKFLOW:


User Starts Chatbot
Collect User Information
Store Data in Google Sheets
Send Data to AI Engine
AI Generates Career Recommendations
Career Selected?
yes
no
Perform Skill Gap Analysis
Ask User to Select Career
Recommend Learning Resources
Generate Comparison Table
Update Skill Progress
Refresh Dashboard & Reports

✅ SEQUENCE DIAGRAM – USER FLOW :



✅ CLASS DIAGRAM – SYSTEM STRUCTURE:



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THANK YOU